Classification Assignment Report

Bishram Acharya

April 28, 2025

Contents

1 Introduction

Breast cancer remains one of the most prevalent and life-threatening diseases affecting women worldwide, with early detection and accurate classification being critical for effective treatment and improved survival rates. The classification of breast tumors-distinguishing between benign and malignant cases-plays a vital role in guiding clinical decisions and patient management. Recent advances in machine learning have enabled the development of automated systems capable of analyzing complex medical data to assist in this classification task. This report presents a breast cancer classification project utilizing machine learning techniques to analyze features extracted from breast cell images.
You can find the project code **here**.

2 Problem Statement

1. Manual classification of breast cancer histopathological images is time-intensive and subject to diagnostic inconsistencies.
2. There is a need for an automated, accurate machine learning model to reliably distinguish between benign and malignant breast tumor images to support clinical decision-making.

3 Objectives

1. To develop, compare, and interpret two machine learning models for accurate classification of benign and malignant breast tumor images from the BreakHis 400x dataset to identify the most effective approach for clinical support.

4 Dataset

The BreakHis 400x dataset is a subset of the Breast Cancer Histopathological Image Classification (BreakHis) database [6], consisting of microscopic images of breast tumor tissue collected from 82 patients.Specifically, the 400x subset contains high-resolution color images (700x460 pixels, RGB) captured at 400 times magnification. It supports both binary and multiclass classification and facilitates benchmarking of diagnostic algorithms in medical imaging research.

Dataset	Benign	Malignant	Total
Train	371	777	1148
Test	176	369	545
Overall	**547**	**1146**	**1693**

Table 1: Distribution of benign and malignant samples in train and test datasets.

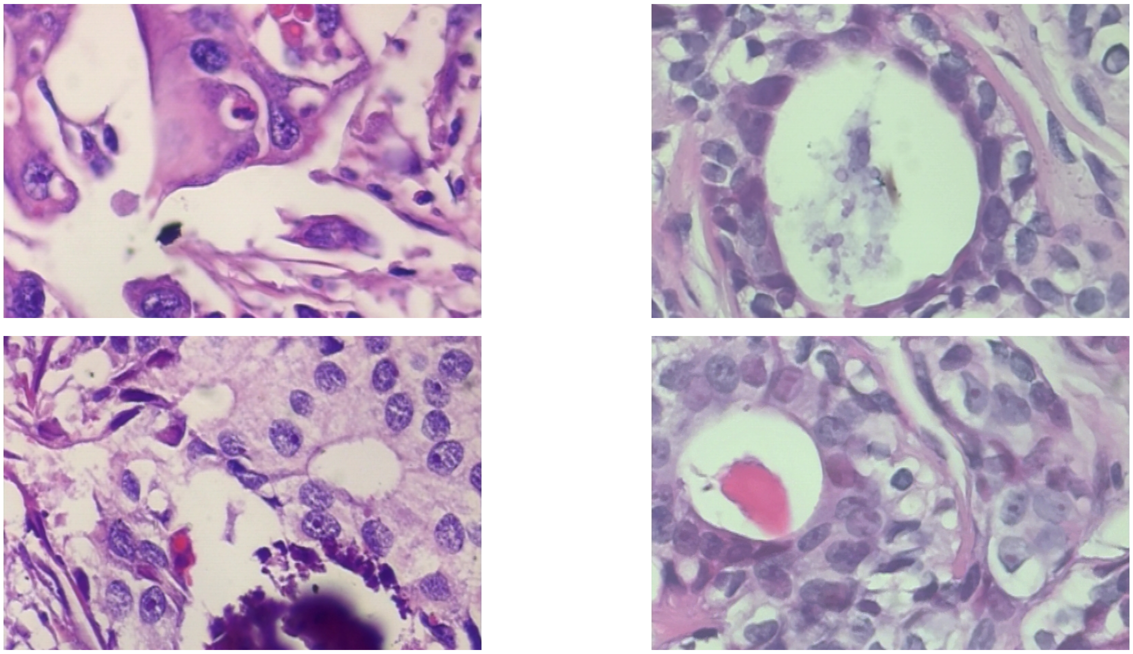

Figure 1: Sample histopathological images of benign (right) and malignant (left) breast cancer tissues. [6]

5 Motivation for my approach

During my search for classification dataset, I came acroos lung cancer(LC25000) dataset [2], which consisted of histopathological images of lung cancer for three different classes and colon cancer for two different classes. A notable work on the lung cancer subset of this dataset was [4] where the authors utilized a hybrid model for classification of lung cancer and achieved promising results.

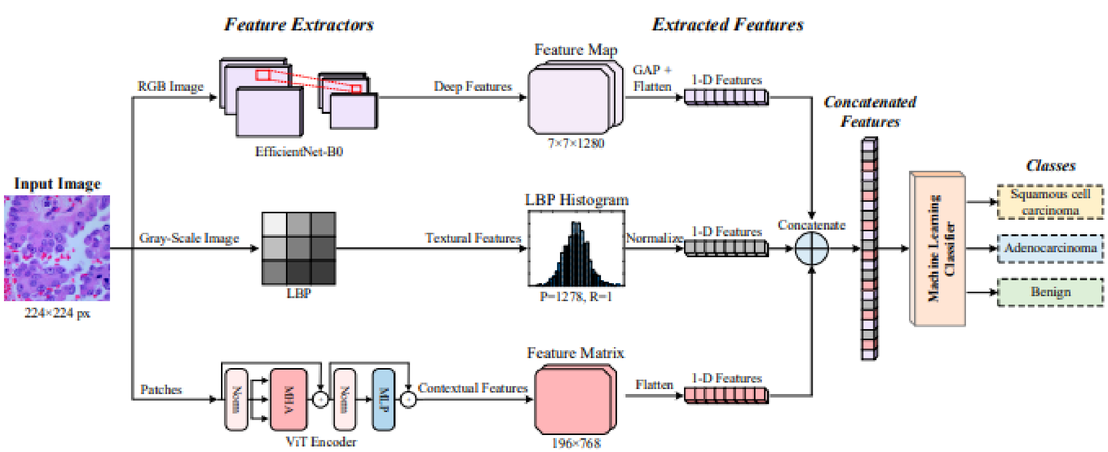

Figure 2: Block diagram of hybrid approach proposed in [4] for lung cancer detection.

Since the nature and texture of histopathological dataset are similar, the working hybrid approach for lung cancer dataset should work on breast cancer dataset. Therefore, I decided to transfer the methodology of [4] into BreakHis 400X dataset and see if I can achieve promising results.

6 Methodology

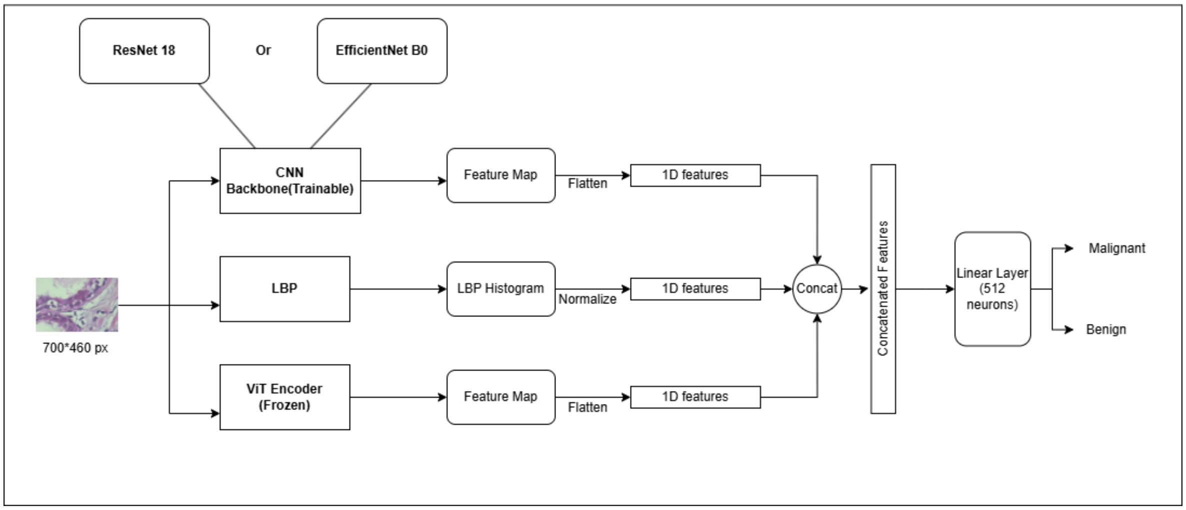

Figure 3: Block diagram of my approach for breast cancer detection.

6.1 EfficientNet-B0

EfficientNet-B0, with 4 million parameters, was selected for its efficiency and effectiveness in image classification tasks. It employs compound scaling to balance network depth, width, and resolution. In breast cancer detection, it helps to automatically learn hierarchical features from histopathological images.

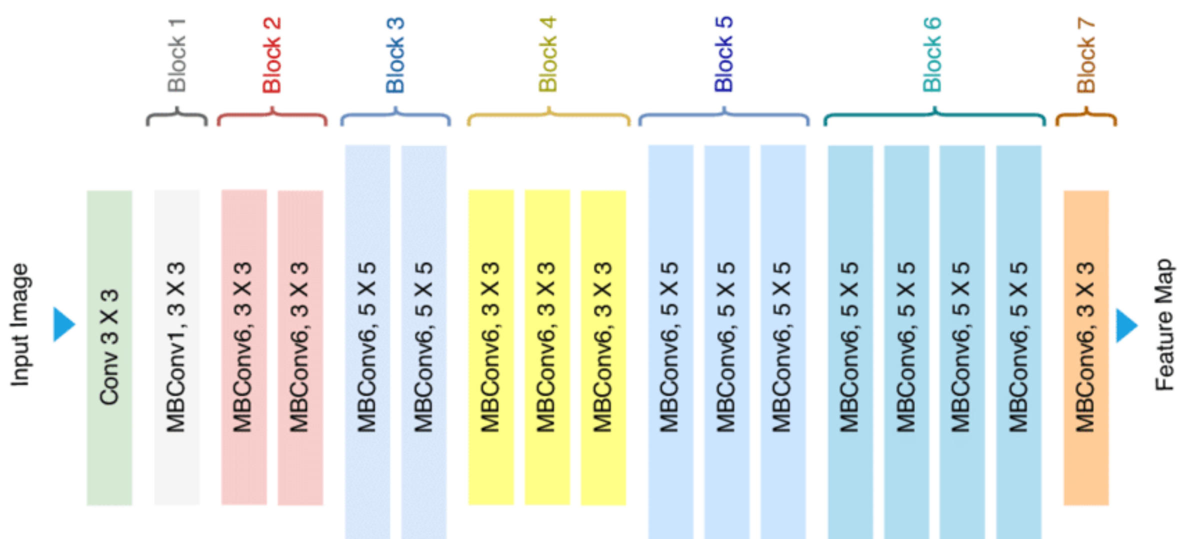

Figure 4: EfficientNet_b0 [1]

6.2 ResNet18

ResNet18, a convolutional neural network with 18 layers, addresses the vanishing gradient problem with residual connections. Its ability to learn intricate image patterns makes it valuable for differentiating subtle variations in breast cancer histology, thus improving diagnostic accuracy.

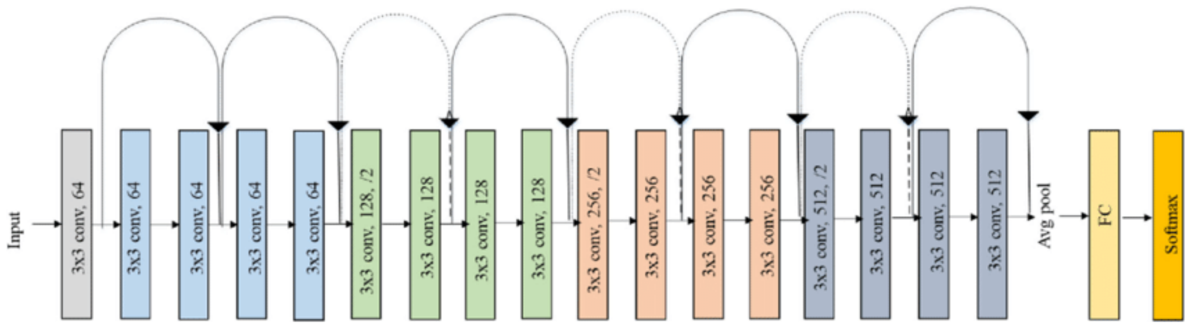

Figure 5: ResNet 18 [5]

6.3 Local Binary Pattern (LBP)

LBP is a texture operator that summarizes local image structures by thresholding pixels based on their neighbors and encoding the result as a binary number. In breast cancer classification, LBP effectively captures textural differences between benign and malignant tissues, providing complementary information to deep learning models and improving overall diagnostic accuracy.

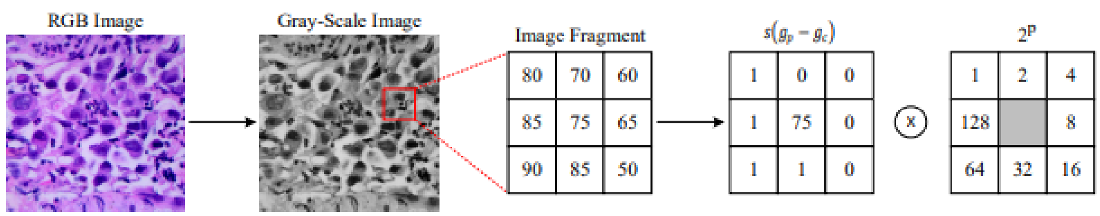

Figure 6: LBP Extraction [4]

6.4 Vision Transformer (ViT) Encoder

The ViT[3] encoder divides images into patches and applies self-attention to learn global relationships. Given the complex spatial arrangements of cells and tissues in histopathological images, ViT's capacity to capture long-range dependencies becomes essential for identifying critical diagnostic features of breast cancer, leading to more accurate classifications.

7 Exploratory Data Analysis (EDA)

7.1 Class Distribution

The BreakHis 400x dataset used in this study exhibits a class imbalance, with 371 benign and 777 malignant images. Figure 7 visualizes the class distribution, highlighting a predominance of malignant samples. This imbalance may influence model performance and will be considered during evaluation.

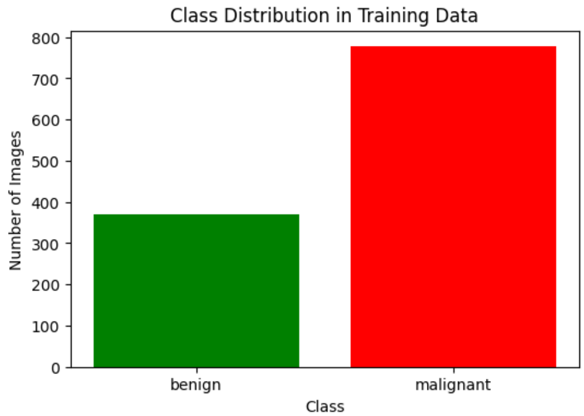

Figure 7: Class distribution of benign and malignant images in the BreakHis 400x dataset.

7.2 Color Channel Analysis

To further understand the dataset, the distributions of the red, green, and blue color channels were examined separately for benign and malignant images (Figures 8, 9, and 10).

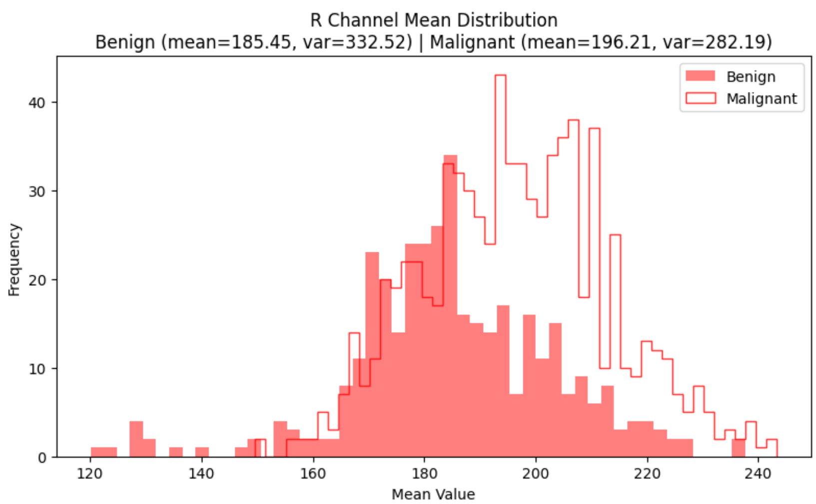

Figure 8: Red channel intensity distribution for benign and malignant samples.

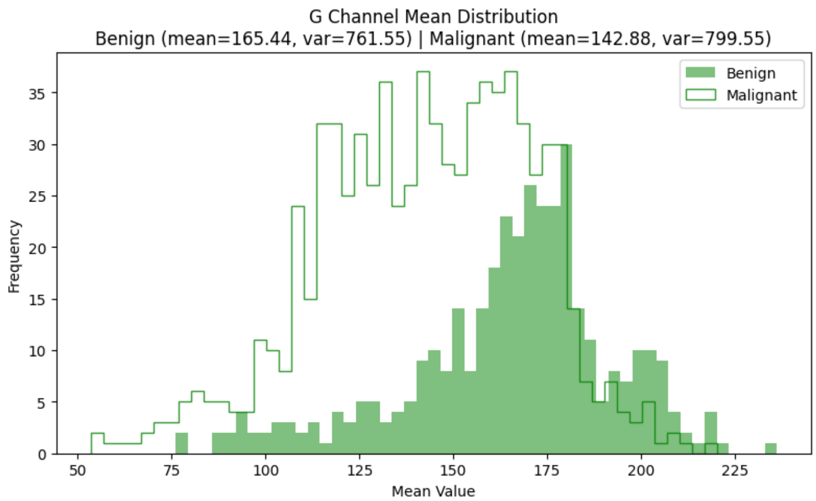

Figure 9: Green channel intensity distribution for benign and malignant samples.

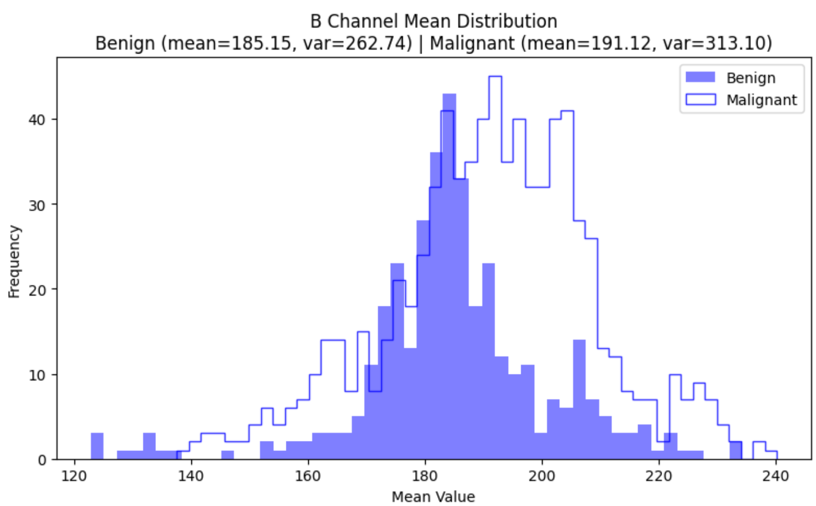

Figure 10: Blue channel intensity distribution for benign and malignant samples.

7.3 Interpretation

Exploratory analysis reveals that malignant tissues tend to be more bluish-reddish and less greenish overall compared to benign tissues. Additionally, the variance of color intensities generally increases in malignant samples (except for a slight decrease in the red channel), indicating higher texture and color irregularities in cancerous tissues. These shifts in color channel distributions suggest potential discriminative features that can be exploited by classification models.

8 Training Setup

The models were trained using the Kaggle environment, which provides a convenient and accessible platform for machine learning experiments. Training was performed on

an NVIDIA Tesla P100 GPU, enabling efficient processing of the high-resolution images from the BreakHis 400x dataset.
The loss function employed was CrossEntropyLoss and adam optimizer was used. Similarly, weight decay was used for regularization purpose and class weights are assigned to loss fn for addressing class imbalance.
The test-train split was roughly 70-30.
The training was configured with the following hyperparameter(batch size, learning rate) settings:
(32, 5e-6)
(32, 5e-5)
(32, 5e-4)
(16, 5e-6)
(16, 5e-5)
(16, 5e-4)
The training graphs are present in Appendix A.

9 Results & Interpretation

9.1 Comparison table

The experiment was run with the best performing configuration (batch size =32 and lr = 5e05) for 10 times and confidence interval was calculated for the evaluation metrics.

Metric	**Accuracy ± CI**	**F1-score ± CI**	**AUC ± CI**	**Precision**	**Recall**
ResNet18	0.9450 ± 0.50%	**0.9600 ± 0.39%**	0.9849 ± 0.0020	0.97	0.97
EfficientNetB0	**0.9580 ± 0.46%**	0.9580 ± 0.46%	**0.9907 ± 0.0016**	**0.98**	**0.98**

Table 2: Comparison of model performance metrics

The result shows that EfficientNetB0 performs slightly better than ResNet18. Although EfficientNetB0 has almost half as less as parameters as ResNet18, it outperforms resnet showing efficiency of the network.

9.2 Confusion matrix

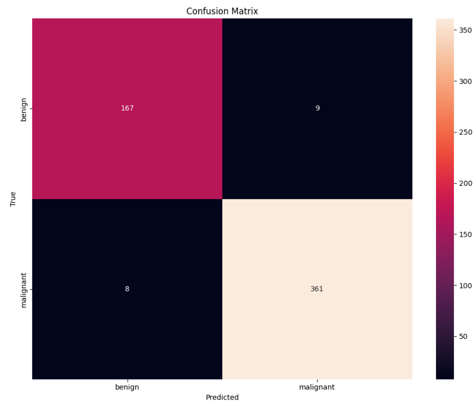

Figure 11: Confusion matrix of the best performing EfficientNet_b0 model

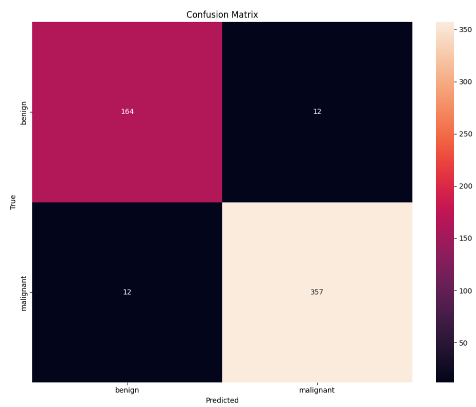

Figure 12: Confusion matrix of the best performing ResNet18 model

9.3 ROC - AUC

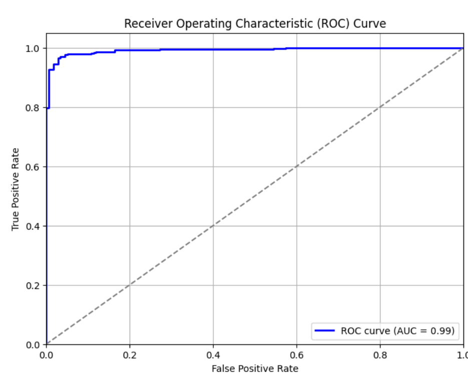

Figure 13: ROC curve of the best performing EfficientNet_b0 model

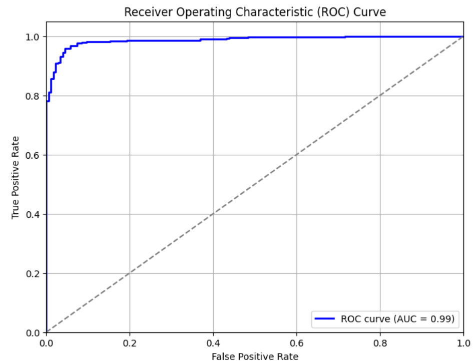

Figure 14: ROC curve of the best performing ResNet18 model

9.4 Class Activation Maps

To investigate how ResNet-18 and EfficienNet_b0 learns and represents information across different layers, I applied Grad-CAM visualizations to an early layer, two intermediate layers, and a final convolutional layer. The results highlight the hierarchical nature of feature extraction in convolutional neural networks.

9.4.1 ResNet18

Correct Prediction

We can observe the following :

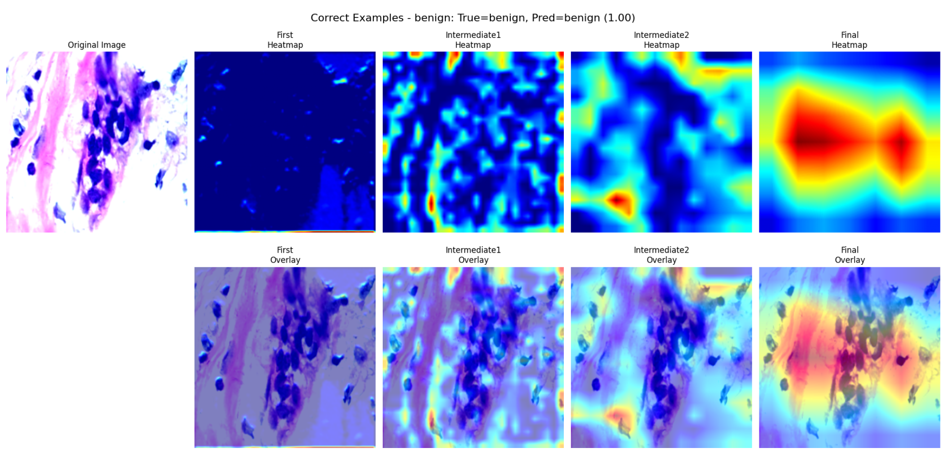

Figure 15: Grad-CAM result on ResNet18 - Correctly Classified Image

1. Early layer : The activations are diffuse and less spatially precise, suggesting that the network is detecting basic visual primitives without focusing on specific object parts.
2. Intermediate layers : Grad-CAM visualizations show mid-level patterns, such as object parts or specific textures, indicating an increasing abstraction of features.
3. Final Layer: In the final convolutional layers before the fully connected head, the Grad-CAM maps become highly focused and discriminative. Activations concentrate around the most class-specific regions of the image.

Incorrect Predictions

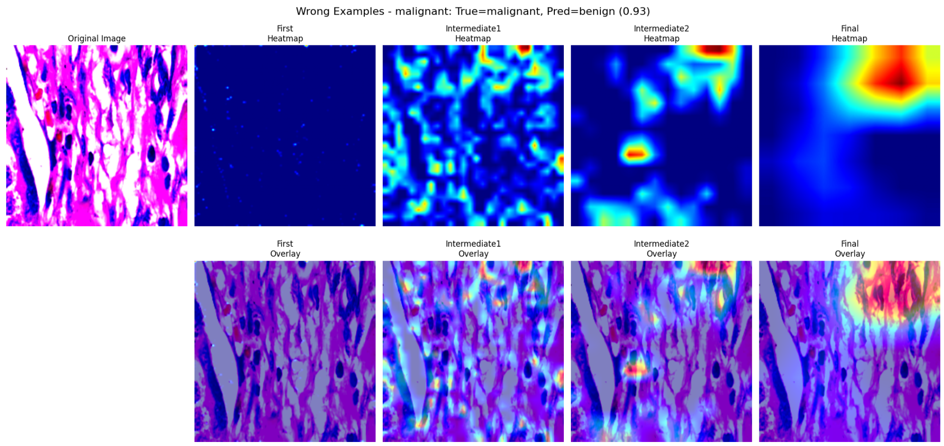

Figure 16: Grad-CAM result on ResNet18 - Incorrectly Classified Image Sample 1

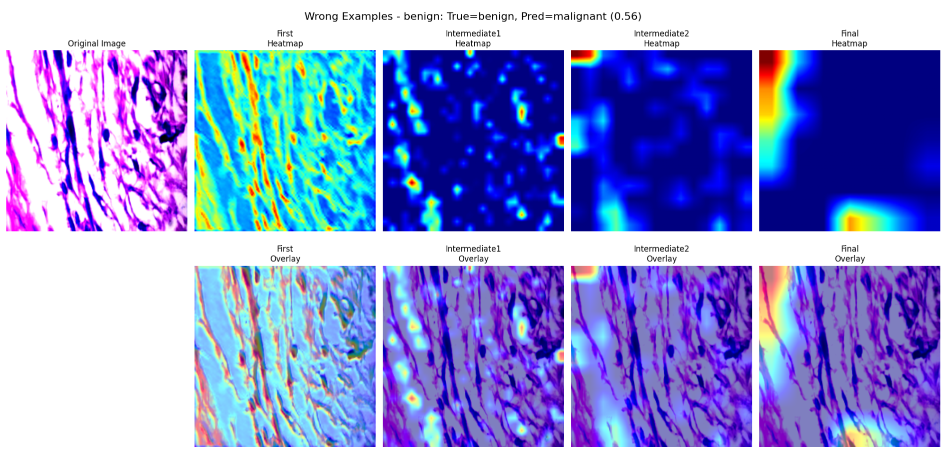

Figure 17: Grad-CAM result on ResNet18 - Incorrectly Classified Image Sample 2

It is observed that when final layer activations are high at very small region at edges and no activation at central regions, such predictions are incorrect. The model struggles when the information is distributed throughout the image as shown in above samples.

9.4.2 EfficientNet_b0

Correct Prediction

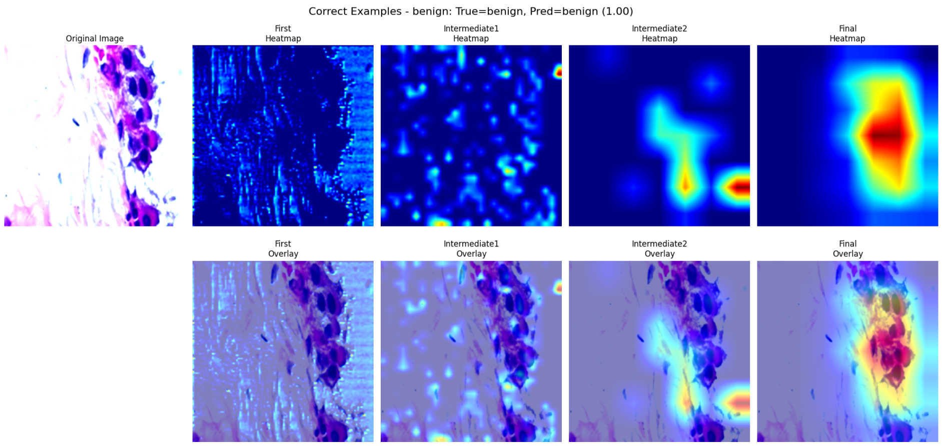

Figure 18: Grad-CAM result on EfficientNet_b0 - Correctly Classified Image

We can see similar GradCAM correctly classified results on EfficientNet_b0 like we see on ResNet.

Incorrect Predictions

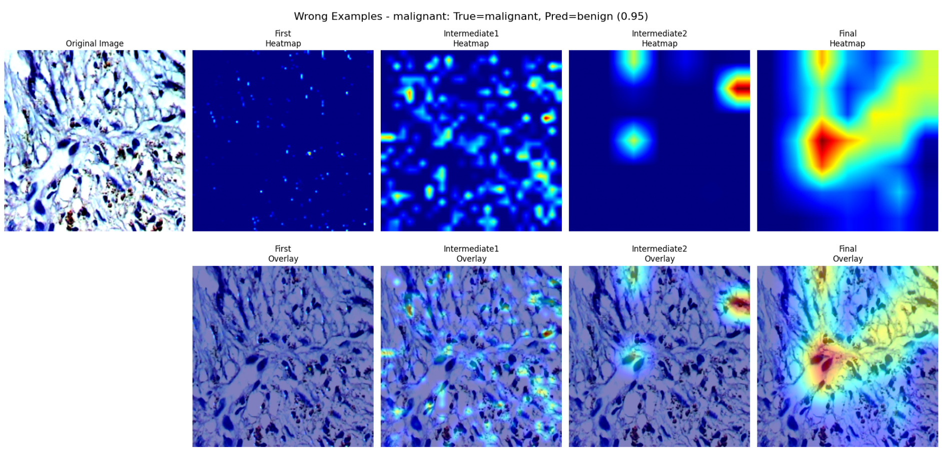

Figure 19: Grad-CAM result on EfficientNet_b0 - Incorrectly Classified Image Sample 1

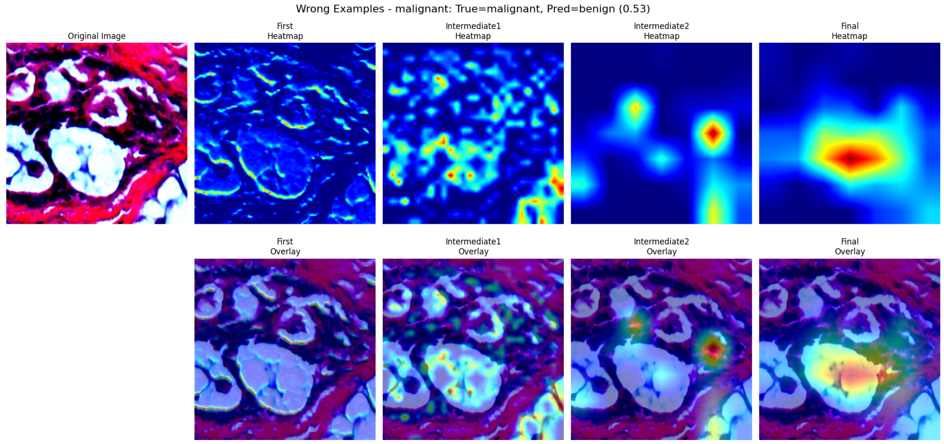

Figure 20: Grad-CAM result on EfficientNet_b0 - Incorrectly Classified Image Sample 2

The EfficientNet model like resnet finds it difficult to classify images where the information is distributed throughout the image. The only difference with resnet is that activations are not at edges but concentrated at center.

10 Conclusion

The choice of model for such a breast cancer classification task should be EfficientNet_b0 because it has much fewer parameters with slighltly better performance than resnet18. In terms of interpretability both models are similar.
The study also shows that methodology from lung cancer classification problem can be transferred to breast cancer classification with great results. This may be because both problem consist of similar nature of images(histopathological).

11 Limitation

One notable limitation of this study is the absence of k-fold cross-validation during model evaluation. Although k-fold validation generally provides a more robust estimate of model performance, it was not employed due to time constraints and computational resource limitations. Instead, a fixed train-test split with a relatively small test size was used. This approach may affect the generalizability and reliability of the reported results, as the model's performance could vary with different data splits.

References

[1] Tashin Ahmed and Noor Sabab. Classification and understanding of cloud structures via satellite images with efficientunet. *SN Computer Science*, 3, 2022.

[2] Andrew A. Borkowski, Marilyn M. Bui, L. Brannon Thomas, Catherine P. Wilson, Lauren A. DeLand, and Stephen M. Mastorides. Lung and colon cancer histopathological image dataset (lc25000). *Journal of Pathology Informatics*, 9(1):34, 2018.

[3] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021.

[4] Oguzhan Katar, Ozal Yildirim, Ru-San Tan, and U Rajendra Acharya. A novel hybrid model for automatic non-small cell lung cancer classification using histopathological images. *Diagnostics*, 14(22), 2024.

[5] Farheen Ramzan, Muhammad Usman Khan, Asim Rehmat, Sajid Iqbal, Tanzila Saba, Amjad Rehman, and Zahid Mehmood. A deep learning approach for automated diagnosis and multi-class classification of alzheimer's disease stages using resting-state fmri and residual neural networks. *Journal of Medical Systems*, 44, 2019.

[6] Fabio A Spanhol, Luiz S Oliveira, Caroline Petitjean, and Laurent Heutte. A dataset for breast cancer histopathological image classification. *IEEE Transactions on Biomedical Engineering*, 63(7):1455–1462, 2016.

A Experiments

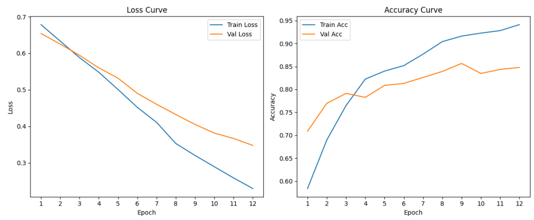

Figure 21: ResNet18 config: (32, 5e-6)

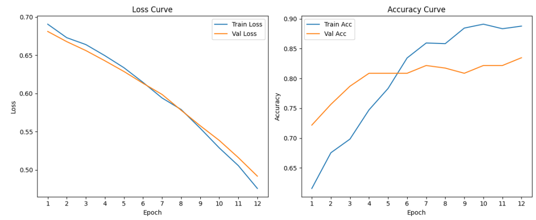

Figure 22: EfficientNet_b0 config: (32, 5e-6)

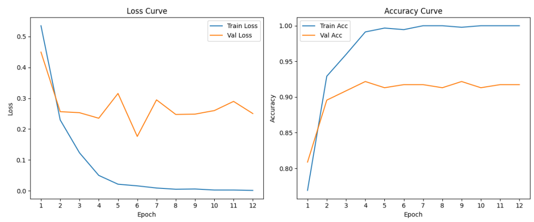

Figure 23: ResNet18 config: (32, 5e-5)

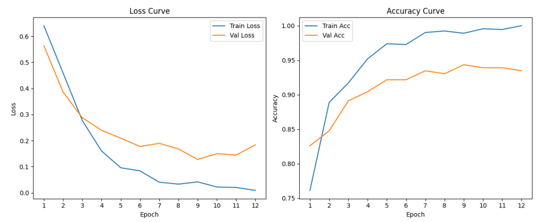

Figure 24: EfficientNet_b0 config: (32, 5e-5)

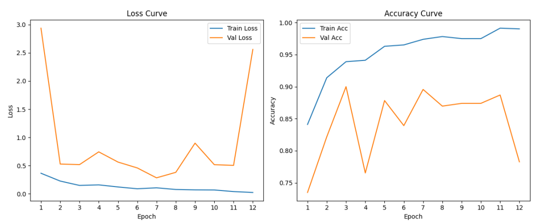

Figure 25: ResNet18 config: (32, 5e-4)

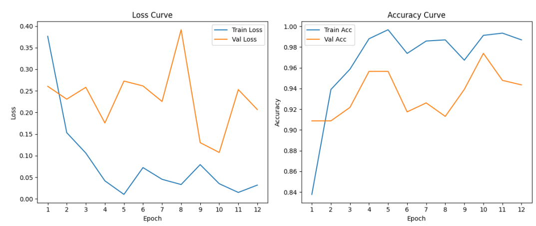

Figure 26: EfficientNet_b0 config: (32, 5e-4)

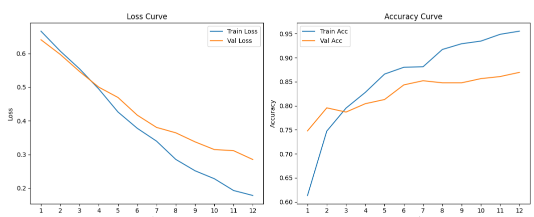

Figure 27: ResNet18 config: (16, 5e-6)

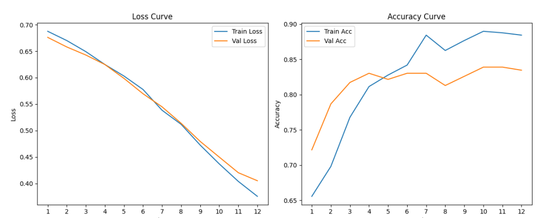

Figure 28: EfficientNet_b0 config: (16, 5e-6)

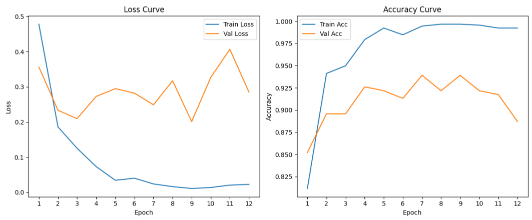

Figure 29: ResNet18 config: (16, 5e-5)

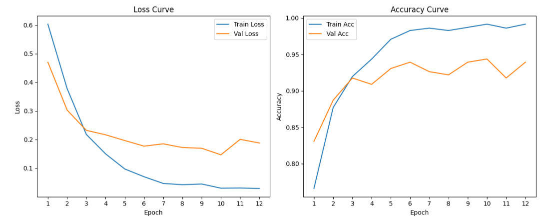

Figure 30: EfficientNet_b0 config: (16, 5e-5)

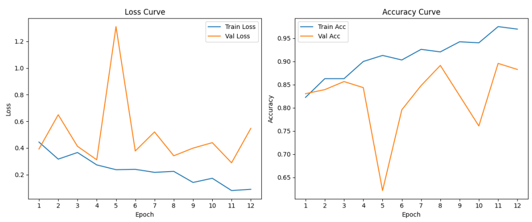

Figure 31: ResNet18 config: (16, 5e-4)

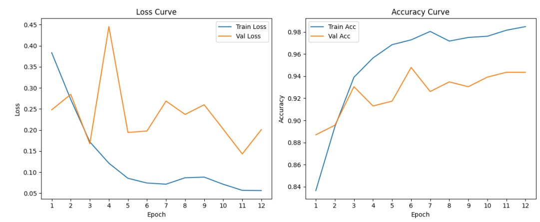

Figure 32: EfficientNet_b0 config: (16, 5e-4)

B Code Snippets

B.1 HybridNet Class

```python
class HybridNet(nn.Module):
    """
    Hybrid classification model that combines CNN backbone, Vision Transformer features,
    and LBP texture features. Supports EfficientNet-B0 or ResNet-18 as backbone.
    """
    def __init__(self, num_classes, lbp_dim=LBP_DIM, vit_dim=VIT_DIM, backbone=BACKBONE_MODEL):
        super().__init__()
        self.backbone_type = backbone

        # Initialize the CNN backbone
        if backbone == 'efficientnet':
            self.cnn = efficientnet_b0(weights=EfficientNet_B0_Weights.IMAGENET1K_V1)
            self.cnn.classifier = nn.Identity()
            self.feature_dim = 1280
        elif backbone == 'resnet':
            self.cnn = resnet18(weights=ResNet18_Weights.IMAGENET1K_V1)
            self.cnn.fc = nn.Identity()
            self.feature_dim = 512
        else:
            raise ValueError(f"Unsupported backbone model: {backbone}. Choose 'efficientnet' or 'resnet'")

```

```python
        # Vision Transformer
        self.vit = ViTModel.from_pretrained(VIT_MODEL)
        for param in self.vit.parameters():
            param.requires_grad = False

        # Combined classifier
        self.classifier = nn.Sequential(
            nn.Linear(self.feature_dim + vit_dim + lbp_dim, 512),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(512, num_classes)
        )

    def forward(self, img_tensor, vit_inputs, lbp_feat):
        """
        Forward pass through the hybrid model.

        Args:
            img_tensor (torch.Tensor): Image tensor for CNN backbone
            vit_inputs (dict): Inputs for Vision Transformer
            lbp_feat (torch.Tensor): LBP texture features

        Returns:
            torch.Tensor: Logits for classification
        """
        # Get CNN features
        cnn_feat = self.cnn(img_tensor)

        # Get ViT features
        with torch.no_grad():
            vit_out = self.vit(**vit_inputs).last_hidden_state[:, 0, :]  # (B, 768)

        # Combine features
        combined = torch.cat([cnn_feat, vit_out, lbp_feat], dim=1)  # (B, total_dim)
        return self.classifier(combined)
```

Listing 1: HybridNet class definition